

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 03 Exploratory Data Analysis

Group Member Contributions

Monepha Mitchell

For this lab, I contributed by completing and reviewing the Python code that demonstrated the bias-variance tradeoff. I trained polynomial regression models with degrees 1, 4, and 15 and compared their prediction lines against the original sine-wave dataset. I also created learning curves for degree 1 and degree 15 models to compare their training and cross-validation scores. After reviewing the graphs, I identified the degree 1 model as underfitting, the degree 4 model as the best fit, and the degree 15 model as overfitting. I completed the knowledge-check questions by explaining how high bias and high variance affected each model. My contribution helped connect the written concepts to visual examples and showed how model complexity affects performance on both training and new data.

Ved Dalwadi

In this module I worked through the bias variance tradeoff lab. I completed all cells and some of the things I did were. Comparing polynomial regression at different degrees to see what a good and bad model will look like in a real scenario. I also Understood the learning curve to see what underfitting, overfitting and good fit models look like to help my and my teammates understanding later on in the course.

Nigel Bontif